Tree Introduction & Terminologies Data Structures

> **Definition:** A **Tree** is a non-linear, hierarchical data structure consisting of a collection of nodes connected by directed or undirected edges, such that there exists exactly one path between any two nodes and no cycles are formed.

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1. Detailed Technical Explanation

Unlike linear data structures (Arrays, Linked Lists, Stacks, Queues) where elements are stored sequentially, trees organize data hierarchically.

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Core Tree Terminologies:

1. **Root:** The topmost node in a tree with no parent (Node `A`).

2. **Ed

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2. Memory Keywords & Mathematical Relations

- **Non-linear Hierarchy:** Parent-child recursive relationship.

- **N

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3. Must-Write Points for Exams

- A tree with `N` nodes always has exactly `N - 1` edges.

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4. Quick Recall Flow

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